Moges Tesema Mekonen

Software Engineer

moges2421@gmail.com | +251974263715 | Adama, ETH | moges | moges

Education

B.Sc in Software Engineering, *Adama Science and Technology University* — 10/2020 – 06/2024, Adama, Ethiopia

- Relevant Course Work: Probability and Statistics, OOP, Operating Systems, Computer Networking, Database System, System Security, Software Architecture and Design. Distributed System, NLP, Computer Vision, Machine Learning.

Competitive programming Education, *Africa to Silicon Valley* — 07/2024 – 11/2024, Adama, Ethiopia

- Relevant Coursework: Data structures and Algorithms (Graph and Tree Algorithms, Dynamic Programming)

Professional Experience

AI/Backend Engineer, *Ethronics-Robotics and Automation Systems* — 12/2024 – 07/2025, Adama, Ethiopia

Developed a distributed **FastAPI** and **PostgreSQL** microservice system for sub-**100ms** toxicity detection, optimizing the Detoxify **NLP** model via quantization to cut memory footprint by **40%** while preserving **96.5%** accuracy. Led a 4-member **Agile** team using **Jira**. Architected asynchronous Celery workflows and a Docker-based **CI/CD pipeline**, utilizing **Redis** caching to reduce redundant model calls by **27%.** Integrated services across Neon (database), Render (API hosting), Railway (background workers), and Hugging Face (model registry).

Projects

Scalable RESTful E-Commerce API, — 03/2023 – 09/2023

Scalable RESTful Architecture with Distributed Task Queues & Caching

Architected a **Django/DR**F and **PostgreSQL** backend sustaining **252+ RPS** at less than **200ms latency** via optimized query patterns. Improved response times by **45%** through **Redis** caching and N+1 resolution, while cutting checkout lag by **250**ms by offloading all background tasks to **Celery**. Guaranteed production stability with **95% Pytest** coverage and **Docker** environment parity for zero regression deployments.

FoodVision, *Deep Learning CNN Transfer Learning Image Classification* — 01/2024 – 06/2024

Developed CNN-based food image classifier and fine-tuned EfficientNetB0 on ImageNet-101. Achieved **91%** training and **93%** validation accuracy with quantization-aware training. Containerized with Docker and deployed using **FastAPI** and **ONNX**. Load-tested with **Locust**, supporting **951 RPS** and **4K** concurrent connections under simulated traffic.

Skills

Languages and Framework

Python, JavaScript, TypeScript, Django, FastAPI, React, Flask, TensorFlow, PyTorch, Keras, NumPy.

Database and Tools

PostgreSQL, MySQL, MongoDB, Locust, Pytest, Git, GitHub Actions, Jira, ONNX